In [1]:
```
import pandas as pd
```

In [2]:
```
# TASK 1: Movies CSV - Using loc()

movies = pd.read_csv("movies.csv")
```

In [3]:
```
print(movies.loc[:, ["title", "rating"]])
```

```
                                  title  rating
0                     Avengers: Endgame     8.4
1                                Avatar     7.8
2                               Titanic     7.9
3                       The Dark Knight     9.0
4                             Inception     8.8
5                          Interstellar     8.7
6                                 Joker     8.3
7                           Toy Story 4     7.7
8                         Black Panther     7.3
9                             Frozen II     6.8
10              Spider-Man: No Way Home     8.2
11                    Top Gun: Maverick     8.2
12                        The Lion King     6.8
13                       Jurassic World     6.9
14  Harry Potter and the Deathly Hallows    8.1
15                The Lord of the Rings     8.9
16                             Iron Man     7.9
17                       Doctor Strange     7.5
18              Guardians of the Galaxy     8.0
19                         Wonder Woman     7.4
```

In [4]:
```
# TASK 2: Flipkart Products - Using iloc()

flipkart_products = pd.DataFrame({"product_name": [
"Samsung Galaxy S24","Apple iPhone 15","Sony Headphones","Dell Laptop","Boat Smartwatch","HP Keyboard",
"Logitech Mouse","OnePlus Nord","JBL Speaker","Canon Camera","Mi Power Bank","Lenovo Tablet"],
"category": ["Mobile","Mobile","Electronics","Laptop","Smartwatch","Computer Accessories","Computer Accessories","Mobile",
"Audio","Camera","Accessories","Tablet"],
"price": [64999,69999,7999,54999,2999,1499,999,24999,4999,45999,1999,18999]})
```

```
print("\nTASK 2 - First 10 Flipkart Products")
print(flipkart_products.iloc[:10, :])
```

```
TASK 2 - First 10 Flipkart Products
         product_name              category  price
0  Samsung Galaxy S24                Mobile  64999
1     Apple iPhone 15                Mobile  69999
2     Sony Headphones           Electronics   7999
3         Dell Laptop                Laptop  54999
4     Boat Smartwatch            Smartwatch   2999
5         HP Keyboard  Computer Accessories   1499
6      Logitech Mouse  Computer Accessories    999
7        OnePlus Nord                Mobile  24999
8          JBL Speaker                 Audio   4999
9        Canon Camera                Camera  45999
```

In [6]:

```
# TASK 3: Zomato Food Orders - Boolean Indexing

zomato_orders = pd.DataFrame({"restaurant": ["Domino's Pizza","Biryani Blues","McDonald's","The Good Bowl","Behrouz Biryani",
"Pizza Hut","Burger King"],
"order_amount": [
450,650,320,720,850,550,400],
"delivery_time": [35,45,30,40,50,38,32]})

high_value_orders = zomato_orders[zomato_orders["order_amount"] > 500]

print("\nTASK 3 - Zomato Orders Above 500")
print(high_value_orders)
```

```
TASK 3 - Zomato Orders Above 500
        restaurant  order_amount  delivery_time
1    Biryani Blues           650             45
3    The Good Bowl           720             40
4  Behrouz Biryani           850             50
5        Pizza Hut           550             38
```

In [7]:

```
# TASK 4: Spotify Songs - Using query()
spotify_songs = pd.DataFrame({
    "song": [
        "Blinding Lights",
        "Shape of You",
```

```
        "Levitating",
        "Perfect",
        "Stay",
        "Believer",
        "As It Was"
    ],
    "artist": [
        "The Weeknd",
        "Ed Sheeran",
        "Dua Lipa",
        "Ed Sheeran",
        "The Kid LAROI",
        "Imagine Dragons",
        "Harry Styles" ],
        "streams": [
        4500000,
        3800000,
        1500000,
        900000,
        2200000,
        1800000,
        1200000
    ],
    "duration": [
        200,
        233,
        203,
        263,
        141,
        204,
        167
    ]
})
```

In [8]:
```
selected_songs = spotify_songs.query("streams > 1000000 and duration < 180")
```

In [9]:
```
print("\nTASK 4 - Spotify Songs")
print(selected_songs)
```

```
TASK 4 - Spotify Songs
        song         artist  streams  duration
4       Stay  The Kid LAROI  2200000       141
6  As It Was   Harry Styles  1200000       167
```

In [10]:
```
# TASK 5: IPL Matches - Boolean Indexing
ipl_matches = pd.DataFrame({"team1": ["Mumbai Indians","Chennai Super Kings","Royal Challengers Bengaluru",
"Rajasthan Royals","Kolkata Knight Riders","Delhi Capitals"],
"team2": ["Chennai Super Kings","Kolkata Knight Riders","Delhi Capitals","Mumbai Indians","Punjab Kings","Sunrisers Hyderabad"
"winner": ["Mumbai Indians","Chennai Super Kings","Royal Challengers Bengaluru","Mumbai Indians","Kolkata Knight Riders","Delh
"total_runs": [195,210,175,220,188,205]})

filtered_matches = ipl_matches[(ipl_matches["total_runs"] >= 180) & (ipl_matches["total_runs"] <= 220)]

print("\nTASK 5 - IPL Matches Between 180 and 220 Runs")
print(filtered_matches)
```

```
TASK 5 - IPL Matches Between 180 and 220 Runs
                   team1                  team2                 winner  \
0         Mumbai Indians    Chennai Super Kings         Mumbai Indians
1    Chennai Super Kings  Kolkata Knight Riders    Chennai Super Kings
3       Rajasthan Royals         Mumbai Indians         Mumbai Indians
4  Kolkata Knight Riders           Punjab Kings  Kolkata Knight Riders
5         Delhi Capitals    Sunrisers Hyderabad         Delhi Capitals

   total_runs
0         195
1         210
3         220
4         188
5         205
```

In [ ]: